

A Comprehensive Analysis of Diabetic Complications and Advances in Management Strategies

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Diabetes mellitus, particularly type 2 diabetes mellitus (T2DM), is a pervasive chronic disease that affects millions of people worldwide. It predisposes individuals to a range of severe microvascular and macrovascular complications, which drastically impact the patient's quality of life and increase mortality rates owing to various comorbidities. This extensive review explores the intricate pathophysiology underlying diabetic complications, focusing on key mechanisms, such as atherosclerosis, insulin resistance, chronic inflammation, and endothelial dysfunction. It also highlights recent therapeutic advancements, including the introduction of SGLT2 inhibitors and GLP-1 receptor agonists, which provide benefits beyond glycemic control and offer cardiovascular and renal protection. Furthermore, the future position of SGLT2 inhibitors and GLP-1 receptor agonists in terms of the prevention of diabetes and macrovascular diseases will be discussed. Considering the differences in insulin secretion capacity between Western and Asian patients, including Japanese patients, we propose a treatment strategy for high-quality diabetes in Japan.

Key words: Diabetes, Macrovascular, Complications, SGLT2, GLP-1

1. Introduction

Diabetes mellitus (DM) is characterized by chronic hyperglycemia, primarily resulting from insufficient insulin secretion and/or insulin resistance. The pathophysiological mechanisms underlying diabetes are complex and involve various metabolic, inflammatory, and hormonal pathways. According to the International Diabetes Federation, the global prevalence of diabetes has increased sharply over the past 2 decades, with projections indicating that over 700 million individuals will be affected by 2030¹⁾. This alarming increase is attributed to lifestyle factors, including unhealthy diets, physical inactivity, obesity, and genetic predispositions.

The complications associated with diabetes are extensive and can be categorized into microvascular and macrovascular. Microvascular complications,

including diabetic retinopathy, nephropathy, and neuropathy, primarily affect the small blood vessels, leading to significant tissue damage. Macrovascular complications involve large arteries and are the leading cause of mortality among patients with diabetes, including cardiovascular disease (CVD), cerebrovascular disease, and peripheral arterial disease (PAD)²⁻⁴⁾.

Recent pharmacological advancements, such as sodium-glucose cotransporter-2 (SGLT2) inhibitors and glucagon-like peptide-1 (GLP-1) receptor agonists, have shifted the treatment paradigm for diabetes management. These agents not only improve glycemic control but also exhibit cardioprotective and nephroprotective effects, reducing the burden of diabetic complications⁵⁻⁶⁾.

This review delves into the pathophysiology of these complications, examines the current and

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Received: November 26, 2024 Accepted for publication: December 11, 2024

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emerging therapeutic strategies, and provides a roadmap for future research.

2. Difference between Microvascular and Macrovascular Complications due to Atherosclerosis

DM is a chronic disease characterized by systemic vascular damage caused by hyperglycemia, which leads to serious complications in both the microvascular and macrovascular systems. These complications significantly reduce the quality of life of patients with diabetes and have a significant impact on their prognosis. This section discusses differences between microvascular and macrovascular complications.

Microvascular complications include retinopathy, nephropathy, and neuropathy, which occur when hyperglycemia directly affects small blood vessels, such as capillaries and small arteries. Conversely, macrovascular complications include coronary artery disease (CAD), cerebrovascular disease, and PAD, which occur when hyperglycemia promotes atherosclerosis, resulting in damage to the macrovasculature. These complications significantly affect the mortality rate of patients with diabetes.

At first glance, microvascular and macrovascular complications may be viewed as independent pathologies, but they are in fact closely related. For example, diabetic nephropathy (DN) progression increases the risk of atherosclerosis, contributing to the development of CAD and cerebrovascular diseases. In addition, inadequate glycemic control has been shown to synergistically increase both the risks.

Because microvascular and macrovascular complications have different pathologies, treatment strategies differ; however, they are interrelated and require comprehensive risk management. After discussing microvascular and macrovascular complications in detail, we will consider treatment strategies.

3. Microvascular Complications

Microvascular complications in diabetes arise from prolonged exposure to hyperglycemia, which leads to endothelial dysfunction, basement membrane thickening, and impaired microcirculation. These changes disrupt the function of the small blood vessels, causing damage to critical organs and systems.

3.1 Diabetic Retinopathy (DR)

DR is the most common microvascular complication and a leading cause of vision loss globally. It affects approximately one-third of diabetic

patients, and its prevalence increases with disease duration and poor glycemic control⁷⁾.

•Pathophysiological Mechanisms

◦Hyperglycemia and oxidative stress: Chronic hyperglycemia induces oxidative stress through the production of reactive oxygen species (ROS). Elevated ROS levels damage the retinal endothelial cells, disrupt tight junctions, and increase vascular permeability. This results in breakdown of the blood-retinal barrier, leading to macular edema and retinal hemorrhaging.

◦Activation of the polyol pathway: The polyol pathway becomes hyperactive under diabetic conditions. Glucose is converted to sorbitol by aldose reductase, and sorbitol accumulation within the retinal cells causes osmotic stress and cellular dysfunction. This contributes to capillary basement membrane thickening and pericyte loss, which are the hallmarks of early DR.

◦Advanced glycation end-products (AGEs) and inflammatory response: AGEs are formed as a result of chronic hyperglycemia and interact with their receptor (RAGE) on endothelial cells, triggering pro-inflammatory signaling pathways. This interaction promotes the release of cytokines, such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), further exacerbating vascular damage and leading to neovascularization observed in proliferative DR.

◦Vascular endothelial growth factor (VEGF) overexpression and neovascularization: Hypoxia in retinal tissue due to impaired blood flow stimulates the upregulation of VEGF, which induces the formation of new blood vessels. However, these neovessels are structurally weak and prone to leakage, resulting in vitreous hemorrhaging and potential retinal detachment.

•The Clinical Presentation and Diagnosis

Non-proliferative DR (NPDR) is characterized by microaneurysms, intraretinal hemorrhaging, and cotton-wool spots, indicating early retinal ischemia. Proliferative DR (PDR) is characterized by neovascularization, vitreous hemorrhaging, and tractional retinal detachment, which can lead to severe vision loss if left untreated.

Fundoscopic examinations and optical coherence tomography are essential tools for diagnosing and monitoring the progression of DR. Fluorescein angiography can also be used to assess retinal blood flow abnormalities.

•Current and Emerging Treatments (Fig. 1)

•Anti-VEGF therapy: Agents such as ranibizumab,

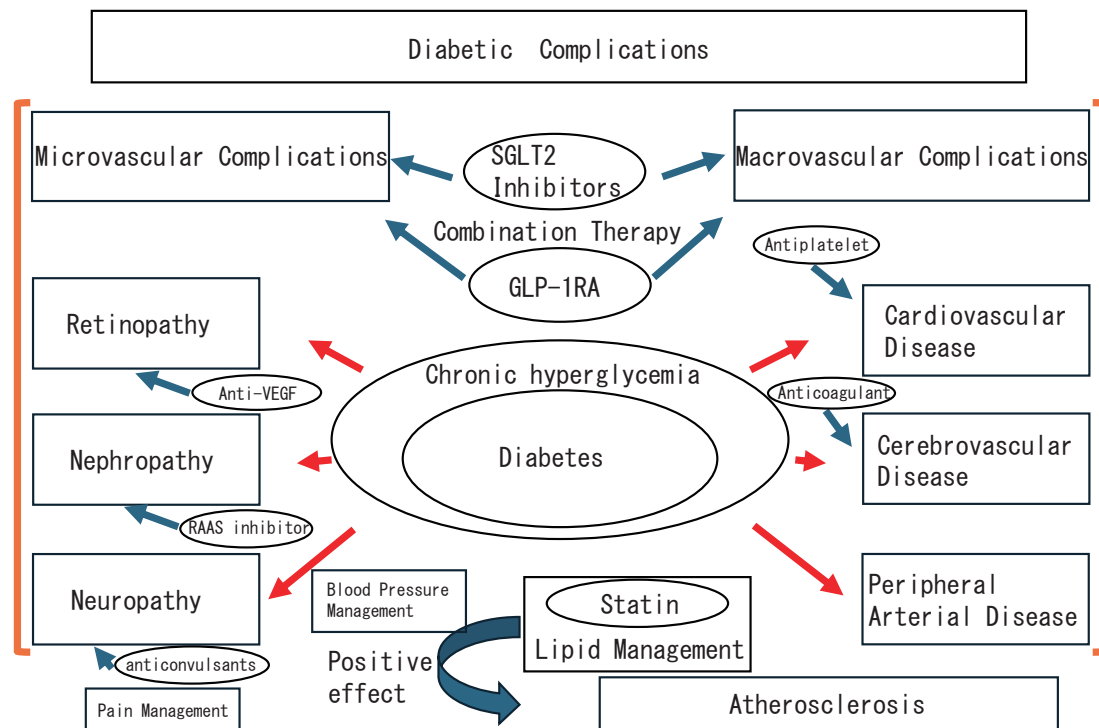


Fig. 1. The treatment strategies for diabetic complications are illustrated

Diabetic complications are categorized into microvascular and macrovascular complications. Microvascular complications include retinopathy, nephropathy, and neuropathy, while macrovascular complications include cardiovascular disease, cerebrovascular disease, and peripheral arterial disease. The red arrows indicate adverse effects, and the blue arrows represent therapeutic effects. Although specific therapeutic agents are available for each complication, SGLT2 inhibitors and GLP-1 receptor agonists have shown therapeutic potential for both microvascular and macrovascular complications. SGLT2 inhibitors, sodium-glucose cotransporter-2 inhibitors; GLP-1RA: glucagon-like peptide-1 receptor agonists; Anti-VEGF: anti-vascular endothelial growth factor; RAAS inhibitor, renin-angiotensin-aldosterone system inhibitor.

bevacizumab, and aflibercept inhibit VEGF activity, reduce macular edema, and halt neovascularization progression. These therapies have become the cornerstone of diabetic macular edema (DME) treatment.

- Corticosteroid injections: Intravitreal corticosteroids, such as dexamethasone implants, are used in cases resistant to anti-VEGF therapy. They reduce inflammation and vascular permeability but carry a risk of increased intraocular pressure and cataract formation.

- SGLT2 inhibitors: Recent studies have suggested that SGLT2 inhibitors may exert protective effects on the retina by reducing oxidative stress and inflammation, highlighting a potential new therapeutic avenue⁸.

- Relationship between DR and Macrovascular Complications:

DR is an indicator of capillary damage caused by high blood sugar levels. Its progression is closely related to chronic inflammation and increased oxidative stress in the entire vascular system, and it has

been suggested to increase the risk of macrovascular lesions in diabetic patients. In particular, severe DR is associated with the progression of atherosclerotic CVD (ASCVD) and can be a predictor of major vascular complications such as CAD, cerebrovascular disease, and PAD⁹.

In fact, the UKPDS and other observational studies have reported that patients with severe DR have significantly higher rates of cardiovascular events and mortality than mild DR. In these patients, atherosclerotic lesions are prominent, and common risk factors, such as hypertension, dyslipidemia, and an impaired renal function, are thought to be involved¹⁰. In addition, it has been confirmed that the risk of cardiovascular events is further increased in patients with concurrent nephropathy, suggesting that DR is intricately related to DN and large-vessel lesions¹¹.

DR also a significant indicator of systemic endothelial dysfunction. Endothelial dysfunction promotes atherosclerosis progression and causes blood flow disorders in the coronary and cerebral arteries.

Chronic hyperglycemia reduces the production of nitric oxide (NO) and decreases vasodilation, creating a vicious cycle that worsens microvascular and macrovascular lesions¹².

These findings suggest that DR is not just a local retinal disease but is also an important indicator of systemic vascular risk. Therefore, its early diagnosis and appropriate management play an important role in the prevention of systemic vascular complications. This is a promising comprehensive treatment strategy for both DR and large-vessel complications.

3.2 DN

DN is a leading cause of chronic kidney disease (CKD) and end-stage renal disease (ESRD). It affects approximately 30%-40% of patients with diabetes and is characterized by progressive albuminuria, a declining glomerular filtration rate (GFR), and the development of glomerulosclerosis¹¹.

•Pathophysiological Mechanisms

◦Glomerular hyperfiltration: Early in the disease course, hyperglycemia leads to glomerular hyperfiltration, which increases the intraglomerular pressure. This state of hyperfiltration damages glomerular capillaries, causing podocyte injury and disruption of the filtration barrier.

◦Renal hypertrophy and inflammation: Chronic hyperglycemia induces renal hypertrophy and mesangial matrix expansion. This is mediated by pro-inflammatory cytokines, such as transforming growth factor- β , which promote extracellular matrix accumulation and fibrosis, leading to glomerulosclerosis.

◦AGEs: AGEs accumulate in the kidneys owing to prolonged hyperglycemia. AGEs interact with RAGE receptors in renal cells, triggering oxidative stress and inflammation. This process accelerates nephropathy progression by promoting tubulointerstitial fibrosis and glomerular damage¹³.

◦Dysregulation of the renin-angiotensin-aldosterone system (RAAS): Hyperglycemia activates the RAAS, increasing angiotensin II levels, which contributes to vasoconstriction, increased glomerular pressure, and subsequent kidney damage. RAAS activation is a key factor in the progression of DN¹⁴.

•The Clinical Presentation and Diagnosis

DN is typically diagnosed based on the presence of albuminuria and a reduced GFR.

◦Microalbuminuria: Early-stage nephropathy is marked by the presence of microalbuminuria (30-300 mg/day), which indicates initial glomerular damage.

◦Macroalbuminuria: As nephropathy progresses,

macroalbuminuria (> 300 mg/day) develops, signaling advanced kidney damage.

◦Declining GFR: A progressive decline in the GFR is observed, leading to CKD and ESRD if left untreated.

•Treatment Strategies

◦RAAS inhibition: Angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers are first-line treatments for DN. They reduce the intraglomerular pressure and slow the progression of albuminuria.

◦SGLT2 inhibitors: These agents lower blood glucose and intraglomerular pressure while reducing albuminuria and delaying CKD progression^{15, 16}. In the EMPA-REG OUTCOME, DECLARE-TIMI, and CANVAS trials, many cases involved patients with an eGFR ≥ 60 and a reduced prevalence of overt nephropathy; however, these studies demonstrated a reduction in end-stage kidney events. Conversely, subsequent trials, such as CREDENCE, DAPA-CKD, and EMPA-KIDNEY, targeted patients with overt nephropathy and a reduced eGFR, also achieving a reduction in end-stage kidney events. Considering these findings, SGLT2 inhibitors effectively suppress end-stage kidney events, such as kidney-related mortality and renal replacement therapy, even in cases with advanced nephropathy or progressive overt nephropathy^{5, 17, 18}.

◦GLP-1 receptor agonists: GLP-1 receptor agonists also show potential benefits in reducing albuminuria and improving renal outcomes, possibly owing to their anti-inflammatory and antioxidant properties. Most previous trials on GLP-1 receptor agonists have primarily targeted patients with a preserved eGFR and a normal state to microalbuminuria. While meta-analyses of these trials indicated a reduction in renal events, statistical significance disappeared when progression to overt albuminuria was excluded as an outcome. In other words, these trials did not demonstrate suppression of true end-stage kidney events¹⁹. However, a recently published FLOW trial showed that semaglutide effectively reduced end-stage kidney events in patients with overt nephropathy, similar to SGLT2 inhibitors²⁰. Based on these findings, it is considered that GLP-1 receptor agonists, particularly semaglutide, can suppress end-stage kidney events, even in cases with overt nephropathy, akin to the effects of SGLT2 inhibitors.

3.3 Diabetic Neuropathy

Diabetic neuropathy is a common chronic complication that affects up to 50% of patients with

diabetes. It involves damage to the peripheral and autonomic nerves, leading to sensory deficits, pain, and an increased risk of foot ulcers and amputations²¹).

- Pathophysiological Mechanisms

- Polyol pathway activation: The polyol pathway becomes overactive in diabetic neuropathy. Excess glucose is converted to sorbitol by aldose reductase, which leads to osmotic stress and nerve cell damage.

- Oxidative stress and inflammation: Chronic hyperglycemia increases ROS production and causes oxidative damage to nerve cells. The interaction between AGEs and RAGE further amplifies inflammation and contributes to nerve dysfunction²²).

- Microvascular damage: Hyperglycemia-induced endothelial dysfunction results in an impaired blood flow to the nerves. Reduced NO production leads to vasoconstriction, exacerbating nerve ischemia and contributing to the progression of neuropathy.

- Clinical Manifestations

Diabetic neuropathy can present in various forms, as follows:

- Peripheral neuropathy: Symptoms include numbness, tingling, and burning pain, which typically start in the feet and progress proximally.

- Autonomic neuropathy: Involves dysfunction of the autonomic nervous system, leading to gastroparesis, orthostatic hypotension, and bladder dysfunction.

- Treatment Approaches

- Pain management: First-line treatments include anticonvulsants (e.g. gabapentin and pregabalin) and antidepressants (e.g. duloxetine), which help alleviate neuropathic pain²³).

- Emerging therapies: GLP-1 receptor agonists and SGLT2 inhibitors are being investigated for their potential neuroprotective effects, with promising preliminary results in reducing the progression of neuropathy²⁴).

4. Macrovascular Complications

Macrovascular complications in diabetes are severe and life threatening, affecting large blood vessels and leading to CVD, cerebrovascular disease, and PAD. These complications are the primary cause of mortality in patients with diabetes. Understanding the pathophysiology and developing effective treatment strategies are critical for improving patient outcomes.

4.1 CVD in Diabetes

CVD includes coronary artery disease, myocardial infarction, and heart failure. Diabetic patients have a two- to four-fold increased risk of CVD compared to non-diabetic individuals, primarily due to insulin resistance, dyslipidemia, chronic inflammation, and endothelial dysfunction^{25, 26}).

- Pathophysiological Mechanisms

- Insulin resistance and lipid dysregulation: Insulin resistance in adipose tissue leads to the release of free fatty acids into the bloodstream, increasing triglyceride levels and reducing high-density lipoprotein cholesterol levels. This lipid profile contributes to atherosclerotic plaque formation and increases the risk of CAD.

- Chronic inflammation: Elevated blood glucose levels induce the release of pro-inflammatory cytokines, such as TNF- α and IL-6, which activate the nuclear factor-kappa B pathway. This chronic inflammatory state exacerbates atherosclerosis by promoting the oxidation of low-density lipoprotein (LDL) cholesterol and increasing the recruitment of macrophages to the vascular endothelium, forming foam cells and plaques.

- Oxidative stress and endothelial dysfunction: Hyperglycemia induces oxidative stress and increases ROS production. ROS damage the endothelial lining of blood vessels, impairing vasodilation and increasing vascular stiffness. Endothelial dysfunction is a key early event in atherosclerosis development.

- AGEs and RAGE activation: AGEs accumulate in the arterial walls of diabetic patients and interact with RAGE receptors. This interaction amplifies inflammatory signaling, leading to increased vascular permeability, plaque formation, and arterial stiffening.

- Clinical Evidence from Major Trials

- EMPA-REG OUTCOME: This landmark trial demonstrated that the SGLT2 inhibitor empagliflozin significantly reduced cardiovascular mortality by 38% and the risk of hospitalization for HF by 35%. These findings suggest that SGLT2 inhibitors have cardioprotective effects beyond glycemic control⁵).

- The DAPA-HF Trial: Dapagliflozin, another SGLT2 inhibitor, showed a 25% reduction in the risk of worsening heart failure or cardiovascular death in patients with heart failure, irrespective of diabetes status, highlighting its broad clinical utility²⁷).

- CANVAS Program: Canagliflozin demonstrated a 14% reduction in the risk of major adverse cardiovascular events (MACE), reinforcing the cardiovascular benefits of SGLT2 inhibitors across different patient populations¹⁸).

•Therapeutic Approaches

The Japan Atherosclerosis Society has published the “Guidelines for the Prevention of Atherosclerotic Cardiovascular Diseases 2022” with the aim of preventing atherosclerotic diseases. These guidelines establish lipid management targets for the prevention of atherosclerotic cardiovascular diseases. Lipid management target values are categorized into low-, medium-, and high-risk groups for primary prevention, with the corresponding targets set for secondary prevention.

Patients with diabetes are classified as high-risk, even for primary prevention. In addition, stricter targets are defined if they have complications, such as PAD, microvascular complications (retinopathy, nephropathy, neuropathy), or a smoking habit. For secondary prevention, the presence of diabetes alone warrants stringent targets equivalent to those for conditions such as acute coronary syndrome, familial hypercholesterolemia, CAD, and atherothrombotic stroke²⁸⁾.

- Statin therapy: Statins are recommended for all patients with diabetes at high cardiovascular risk because of their potent LDL cholesterol-lowering effects. Statins reduce the risk of cardiovascular events by stabilizing atherosclerotic plaques and reducing inflammation.

- Antiplatelet agents: Aspirin and clopidogrel, are used to prevent thrombotic events by inhibiting platelet aggregation. In Japanese patients with type 2 diabetes who have no history of ASCVD, there is no evidence of primary prevention of atherosclerotic events, such as myocardial infarction, with low-dose aspirin in the current situation where risk factors for atherosclerosis are being managed²⁹⁾. In contrast, in diabetic patients with a history of CVD, dual antiplatelet therapy is often used to reduce the risk of recurrent myocardial infarction.

- SGLT2 inhibitors and GLP-1 receptor agonists: Both classes of drugs have demonstrated significant cardiovascular benefits, with reductions in heart failure hospitalizations and MACE. These effects are thought to be mediated by mechanisms such as improved cardiac metabolism, reduced oxidative stress, and enhanced endothelial function. SGLT2 inhibitors demonstrate a heart failure suppression effect of over 30%, whereas GLP-1 receptor agonists exhibit a modest suppression effect of approximately 9%, highlighting a difference in their efficacy. Furthermore, SGLT2 inhibitors reportedly showed consistent heart failure suppression effects across all patient groups. However, regarding atherosclerotic events such as myocardial infarction, SGLT2 inhibitors suppress such events in patients with a

history of CVD but do not exhibit the same effect in high-risk patients without prior CVD. This suggests that the effect of SGLT2 inhibitors on suppressing atherosclerotic events is relatively weak³⁰⁾. In contrast, GLP-1 receptor agonists primarily suppress atherosclerotic events, with particularly strong effects on reducing stroke. Unlike SGLT2 inhibitors, their efficacy appears to be independent of the presence or absence of a history of CVD¹⁹⁾. While further research is necessary to consolidate these findings, current evidence suggests that SGLT2 inhibitors may be primarily beneficial for heart failure suppression, whereas GLP-1 receptor agonists may hold greater potential for preventing atherosclerotic diseases, such as myocardial infarction and stroke.

4.2 Cerebrovascular Disease

Diabetes increases the risk of both ischemic and hemorrhagic stroke. Hyperglycemia exacerbates endothelial dysfunction, increases blood coagulability, and accelerates atherosclerotic changes in the cerebral arteries, making patients with diabetes more susceptible to stroke.

•Pathophysiological Mechanisms

- Hyperglycemia-induced endothelial dysfunction: Chronic hyperglycemia impairs endothelial nitric oxide (NO) production, reduces vasodilation, and promotes arterial stiffness. Endothelial dysfunction increases the risk of plaque rupture leading to thromboembolic stroke.

- Enhanced thrombogenesis: Diabetes is associated with increased platelet activity and fibrinogen levels, creating a pro-thrombotic environment. This increases the risk of clot formation and subsequent ischemic stroke.

- Vascular fragility and hemorrhagic stroke: The accumulation of AGEs and prolonged inflammation weakens the integrity of cerebral blood vessels, increasing their susceptibility to rupture, which can result in hemorrhagic stroke.

•Clinical Management

- Anticoagulant therapy: Warfarin and direct oral anticoagulants (DOACs) like apixaban are effective in reducing stroke risk, especially in diabetic patients with atrial fibrillation. These agents inhibit the coagulation cascade and prevent thrombus formation³¹⁾.

- Blood pressure control: Aggressive blood pressure management is critical to prevent stroke in patients with diabetes. Clinical guidelines recommend maintaining a blood pressure below 130/80 mmHg to minimize stroke risk³²⁾.

4.3 PAD

PAD is a common macrovascular complication of diabetes, characterized by narrowed arteries that reduce blood flow to the limbs. Patients with diabetes have a four-fold increased risk of developing PAD, which can lead to critical limb ischemia and amputation if left untreated.

- Pathophysiological Insights

- Microvascular compromise: Hyperglycemia-induced microvascular damage impairs blood flow to the extremities, exacerbating the effects of macrovascular occlusion in PAD.

- Inflammatory pathways: Chronic inflammation driven by hyperglycemia promotes the development of atherosclerotic plaques in the peripheral arteries, contributing to the progression of PAD.

- Diagnosis and Advanced Therapies

- The ankle-brachial index (ABI) is an effective, non-invasive diagnostic tool for PAD. ABI values below 0.9 indicate the presence of PAD, warranting a further evaluation and management.

- Revascularization techniques: Angioplasty, stent placement, and surgical bypass are critical interventions for severe PAD aimed at restoring blood flow and preventing limb loss.

- Emerging pharmacotherapies: SGLT2 inhibitors and anti-inflammatory agents are being explored for their potential benefits in improving blood flow and reducing the risk of adverse PAD outcomes^{33, 34}.

5. Advanced Therapeutic Strategies for Diabetic Complications

Diabetes management has evolved significantly with the advent of novel pharmacotherapies and combination approaches that target not only blood glucose levels, but also cardiovascular and renal protection. The focus has shifted toward therapies that address the underlying pathophysiology of diabetic complications with the aim of reducing morbidity and mortality.

5.1 Combination Therapy Approaches

The combination of SGLT2 inhibitors and GLP-1 receptor agonists has emerged as a promising strategy that offers synergistic benefits in terms of glycemic control, weight reduction, and organ protection.

SGLT2 inhibitors lower blood glucose by enhancing renal glucose excretion, whereas GLP-1 receptor agonists stimulate insulin secretion and inhibit glucagon release. Together, these provide robust glycemic control without the risk of

hypoglycemia.

Clinical trials have demonstrated that combination therapy reduces the risk of MACE, myocardial infarction, and heart failure hospitalizations more effectively than monotherapy alone.

Regarding clinical evidence, the AWARD-10 trial highlighted the efficacy of combining GLP-1 receptor agonists with SGLT2 inhibitors, showing significant reductions in HbA1c levels and greater weight loss than with individual therapies. Patients also experienced improvements in blood pressure and lipid profiles, reducing the overall cardiovascular risk³⁵.

5.2 Emerging Pharmacological Agents

The development of novel pharmacotherapies has focused on targeting multiple metabolic pathways implicated in the progression of diabetic complications.

Tirzepatide, a dual incretin agonist, is a known pharmacological agent. A dual GLP-1 and GIP receptor agonist, it has shown unprecedented benefits in terms of glycemic control and weight reduction. It acts on both incretin pathways, enhancing insulin secretion and reducing appetite, leading to substantial reductions in HbA1c levels and body weight. The ongoing SURPASS-CVOT trial is expected to reveal the cardiovascular event suppression effects of tirzepatide³⁶.

- Anti-Inflammatory Agents

- IL-1 β inhibitors: Chronic inflammation plays a central role in diabetic complications. IL-1 β inhibitors, such as canakinumab, have been explored for their ability to reduce systemic inflammation and subsequently lower cardiovascular event rates in patients³⁷.

6. Conclusion and Future Directions

DM, particularly type 2 diabetes, is a multifactorial disease with widespread complications that affect multiple organ systems. The burden of microvascular and macrovascular complications highlights the need for comprehensive and multifaceted management strategies. The introduction of novel agents, such as SGLT2 inhibitors and GLP-1 receptor agonists, has shifted the focus from merely controlling blood glucose levels to addressing the broader aspects of cardiovascular and renal protection (Fig. 1).

SGLT2 inhibitors and GLP-1 receptor agonists are currently being used in non-diabetic patients. Specifically, they are employed for cardioprotective

and renoprotective purposes as well as for the management of obesity. Evidence concerning the effects of SGLT2 inhibitors and GLP-1 receptor agonists on macrovascular complications suggests that while the mechanisms through which their pharmacological effects directly exert anti-atherosclerotic actions are being examined, their anti-obesity effects may also lead to favorable metabolic changes, which could secondarily help suppress cardiovascular events.

From the perspective of preventing macrovascular complications in diabetes, focusing solely on extreme reductions in blood glucose levels has shown limited effectiveness. Instead, it is crucial to comprehensively control multiple factors that contribute to atherosclerosis. High-quality glycemic control has the potential to effectively manage macrovascular complications. The positive outcomes emerging from the use of SGLT2 inhibitors and GLP-1 receptor agonists in reducing atherosclerotic disease further bolster optimism regarding future applications.

While this review does not address DPP-4 inhibitors, they remain widely used in Japan. A series of studies conducted approximately a decade ago showed that the effects of DPP-4 inhibitors on cardiovascular events were generally neutral. However, from that point onward, the U.S. FDA began to require evidence of non-inferiority regarding cardiovascular events as a condition for the approval of diabetes medications. DPP-4 inhibitors are highly beneficial for achieving high-quality glycemic control. Given the unique pathophysiology of diabetes in Asian populations, including Japanese patients, which is characterized by impaired insulin secretion, these drugs are indispensable. Although their usage has relatively declined in Western countries, drug selection should consider racial differences into account³⁸.

In this context, while being mindful of global trends in diabetes treatment, it is imperative to develop strategies tailored to the unique needs of our country.

Future research should emphasize the long-term safety and effectiveness of combination therapies as well as the exploration of new therapeutic targets that can address the underlying mechanisms of diabetic complications. Personalized medical approaches, guided by genetic and biomarker profiling, hold promise for optimizing treatment regimens and improving patient outcomes. Continued innovation in pharmacotherapy, combined with lifestyle interventions, will be crucial for reducing the global burden of diabetes and its complications.

Acknowledgements

I would like to express my sincere gratitude to Professor Shimano for providing inspiration and ideas for this review. I would also like to thank Professor Yagyu for checking the details of this paper and providing constructive comments.

Notice of grant Support:

None

Conflict of Interest

H. Shimano received scholarship grants from KOWA COMPANY LTD.

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